THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Hydra	Also yeast
2	**(c)** stamen	Anther and filament
3	**(d)** fallopian tube (oviduct)	Zygote formed
4	**(a)** Incorrect — it creates very little variation	DNA copying errors only
5	**(b)** Student A is correct	Many daughter cells
6	**(c)** spore formation	Sporangia
7	**(b)** vegetative propagation	Layering/grafting
8	**(b)** condom	Barrier
9	**(b)** placenta	Villi
10	**(c)** 23	Half

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It passes on the body design and information from parents to offspring.	2
12	Gonorrhoea, syphilis	1+1
13	They add secretions that provide a fluid medium and nutrition to sperms.	2
14	False — self: pollen to stigma of the same flower; cross: pollen to stigma of another flower.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Planaria cut into pieces regrow into complete organisms through specialised cells; in fragmentation (Spirogyra) the organism breaks up into pieces naturally and each grows	2+1
16	Breast size increases, darkening of nipple skin; hair in armpits and genital area; menstruation begins	3
17	Sepal, petal, stamen (anther, filament), pistil (stigma, style, ovary with ovules)	3
	OR: Pollen lands on stigma, a pollen tube grows through the style to the ovule; male germ cell fuses with the female germ cell to form a zygote, which develops into the embryo	3
18	Plants bear flowers and fruits earlier; seedless plants can be grown; offspring are genetically identical to the parent	3
	OR: To control population and maintain the health of mother; barrier methods (condom), oral pills, copper-T, surgery	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) ovaries, oviducts, uterus, cervix, vagina (ii) uterine lining breaks and comes out as menstruation (iii) through the placenta from mother's blood	2+2+1
	OR: (i) testes, scrotum, vas deferens, seminal vesicles, prostate, urethra, penis (ii) sperm formation requires a lower temperature (iii) carries sperms from testis to urethra	3+1+1
20	(i) yeast — budding; Rhizopus — spore formation (ii) sporangia containing spores (iii) thick walls protect them (iv) Hydra	2+1+1+1
	OR: (i) combines DNA from two individuals giving new combinations (ii) restores the original number in the zygote, keeping chromosome number constant (iii) one parent vs two; no gametes vs gametes	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Amoeba	Splits into two
2	**(d)** pistil (carpel)	Stigma, style, ovary
3	**(d)** seed	Ovary → fruit
4	**(a)** Correct	Filament breaks
5	**(a)** Student A is correct	Hibiscus is bisexual
6	**(b)** buds in leaf notches	Vegetative
7	**(a)** urethra	Common passage
8	**(c)** a virus (HIV)	Viral STD
9	**(d)** nine months	Gestation
10	**(d)** zygote	Fertilised egg

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Transfer of pollen grains from the anther to the stigma.	2
12	Layering, grafting (cuttings, tissue culture)	1+1
13	It helps species survive changing environments; some variants may survive drastic changes.	2
14	True — the lining becomes thick and spongy with blood supply each month.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Amoeba can split in any plane; Leishmania (with flagellum) splits in a definite plane (longitudinally)	3
16	Growing new plants from a few cells of the growing tip in a nutrient medium; used for ornamental plants (orchids) and to produce disease-free plants in large numbers	1+2
17	The placenta is a disc embedded in the uterine wall with villi on the embryo side and blood spaces on the mother's side; glucose and oxygen pass to embryo; wastes removed	3
	OR: Barrier — condom; chemical — oral pills; IUCD — copper-T; surgical — vasectomy/tubectomy	3
18	Maintains body design; copying reactions are not absolutely accurate, giving slight variations useful for evolution	2+1
	OR: Hyphae bear sporangia containing spores; spores are released and germinate on moist surfaces into new Rhizopus	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) pollination, pollen tube growth, fertilisation forming zygote (ii) ovule → seed (embryo + cotyledons); ovary → fruit; petals, sepals fall (iii) development of seed into seedling under favourable conditions	2+2+1
	OR: (i) fission (Amoeba), budding (Hydra), regeneration (Planaria) / spore formation (Rhizopus) (ii) single parent, no gametes, rapid multiplication	3+2
20	(i) HIV-AIDS, warts (ii) condom (iii) it disturbs the sex ratio and is unethical (iv) it is placed in the uterus to prevent implantation	2+1+1+1
	OR: (i) facial hair, deeper voice, enlarged genitals, hair in armpits (ii) sperms in testes; eggs in ovaries (iii) monthly breakdown of unused uterine lining, released as blood and mucus	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.